



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Research Analyze and enrich this hash, include any other matching sampl...

Research Report

Classification	TLP:CLEAR
Published	2026-06-25
Version	1.0
Author	Lost Boys Cyber
Report Type	Research Report

TLP:CLEAR | Lost Boys Cyber | Research Analyze and enrich this hash, include any other matching sampl... - Research Report

Published: 2026-06-25 | TLP:CLEAR | Version: 1.0 | Author: Lost Boys Cyber

AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

Table of Contents

1. Executive Summary
2. Intelligence Requirement
3. Source Review and Confidence
4. Key Findings
5. Threat Context
6. Detection and Hunting
7. Recommendations
8. References

Hash Enrichment Research Report:

54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d

1. Executive Summary

Lost Boys Cyber assessed SHA256 `54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d` as a malicious Windows 32-bit portable executable based on public malware-intelligence enrichment. The strongest directly observed public source was Hybrid Analysis / CrowdStrike Falcon Sandbox, which identifies the file as a 107 KiB PE32 GUI executable for Intel i386 with six sections, submitted on 2025-11-27, and gives a malicious verdict with 58% AV detection. Hybrid Analysis did not expose completed sandbox behavior reports or file relationship pivots for the exact hash, so this report treats static reputation, AV-family labels, packer classification, and family-level references as the primary evidence.

The most actionable enrichment is that engines classified the sample as packed/obfuscated and potentially related to the Midgare/Bifrose backdoor ecosystem: ESET labeled it `a variant of Win32/Packed.PrivateEXEProtector.B suspicious application`, Lionic labeled it `Trojan.Win32.Midgare.ljsQ`, Gridinsoft labeled it `Trojan.Heur!.032100A1`, and CrowdStrike Falcon labeled it `Win/malicious\_confidence\_100%`. Microsoft's public Midgare description states that Midgare contains an encrypted copy of Backdoor:Win32/Bifrose and injects the backdoor into `explorer.exe`, while Kaspersky maps Midgare to spyware/trojan behavior and ATT&CK execution and defense-evasion techniques. Because the exact sample was not available locally and public dynamic telemetry was absent, Lost Boys Cyber assesses the hash as malicious with medium confidence and assesses the Midgare/Bifrose relationship as plausible but not proven for this exact binary.

Priority defensive action is to block the hash at EDR, mail, web proxy, and download-control layers; hunt for the hash and adjacent telemetry; and review endpoints for suspicious `explorer.exe` injection, packed unknown PE execution, unexpected outbound traffic from Windows shell processes, and follow-on credential or espionage behavior. If the binary is later obtained, run full static, TRASHcan dynamic, and Ghidra-backed code analysis before publishing a final sample-backed malware-analysis report.

Field	Assessment
Primary observable	SHA256 `54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d`
Observable type	Windows PE32 executable, GUI subsystem, Intel i386
Public verdict	Malicious in Hybrid Analysis / CrowdStrike Falcon Sandbox
Family / cluster	Unknown exact family; possible Midgare/Bifrose-adjacent packed trojan
Confidence	Medium for maliciousness; low-to-medium for exact family attribution
Main risk	Packed Windows malware with possible backdoor/spyware behavior and process-injection tradecraft
Immediate handling	Quarantine/block hash, preserve host telemetry, hunt for packed PE and shell-process anomalies

2. Intelligence Requirement

The requirement was to analyze and enrich a single hash, identify any matching samples, and produce a Lost Boys Cyber research report suitable for the portal renderer. This report answers four operational questions:

Requirement	Finding	Confidence
What is the file?	A 107 KiB PE32 Windows GUI executable for Intel i386, six sections, public submission date 2025-11-27.	High, from Hybrid Analysis page metadata.
Is it malicious?	Public reputation is malicious: CrowdStrike Falcon 100% malicious-confidence label and Hybrid Analysis malicious verdict; MetaDefender exposed 4/26 detections.	Medium-high, because multiple engines flag it, but public dynamic analysis is absent.
What cluster/family does it match?	AV labels point toward packed malware and a possible Midgare/Bifrose relationship; exact family is not fully proven from public data alone.	Low-to-medium.
Are there matching/related samples?	Hybrid Analysis lists no public sandbox submissions, dropped files, parents, children, or bundled-file relationships for the exact hash. Kaspersky lists example MD5s for the Midgare family, but they are family examples, not confirmed siblings of this exact SHA256.	Medium for no public exact relations; low for family-level examples.

Analyst assumptions:

- The hash is a file SHA256, not a URL or certificate hash.
- No malware sample was provided with the portal task, so this is a research/enrichment report, not a sample-backed reverse-engineering report.
- Public-source pages and search snippets are treated as evidence with source limitations; claims that require binary access are stated as hypotheses.
- The report avoids over-claiming runtime behavior for the exact sample where no sandbox behavior report was available.

3. Source Review and Confidence

The enrichment process checked direct hash matches and family/context pivots across public sources. The exact-hash result was sparse: Hybrid Analysis exposed useful metadata and AV information, while Triage returned no public sample page, Manalyzer reported no report for the exact hash, VirusTotal required authenticated/API access or rate-limited UI access, and MalwareBazaar presented an anti-automation browser challenge. These limitations materially reduce confidence in dynamic behavior and sample relationships.

Source	Result	Use in this report	Confidence / limitation
Hybrid Analysis / CrowdStrike Falcon Sandbox	Exact SHA256 page available; malicious verdict; static file metadata; AV summary; no completed sandbox reports or relations.	Primary exact-hash evidence.	Medium-high for metadata and reputation; low for behavior because no sandbox report was exposed.
OPSWAT MetaDefender via Hybrid Analysis modal	Four detections shown: Aurora, Gridinsoft, Lionix, ESET.	AV label comparison and family hypothesis.	Medium; AV labels vary and can conflate packer/family.
Kaspersky Threats: Trojan.Win32.Midgare.ajzz	Family description: trojan/spyware behavior; ATT&CK mappings include execution and virtualization/sandbox-evasion system checks; page lists	Family-level context for Lionix Midgare label.	Medium for generic family context; not proof this exact SHA256 is Midgare.

	example MD5s.		
Microsoft Security Intelligence search result for Trojan:Win32/Midgate.A	Describes Midgate.A as containing encrypted Backdoor:Win32/Bifrose and injecting it into `explorer.exe`.	Process-injection/backdoor hypothesis and hunt logic.	Medium; search result is specific to Midgate.A, not exact SHA256.
Dr.Web VMS search for Packer.PrivateExeProtector.A	Alias table shows PrivateExeProtector packer detections associated with many backdoor, downloader, banker, autorun, and Bifrost/Bifrose-like labels.	Packer-risk context and caution around family attribution.	Low-to-medium; aliases indicate broad packer abuse, not exact sample lineage.
Public exact-hash web search	No durable high-quality exact-hash writeups found beyond Hybrid Analysis.	Supports source limitation statement.	Medium; search indexing is incomplete.

Key confidence judgment: maliciousness is supported by direct public reputation, but family attribution and behavior should remain caveated until the executable is obtained and detonated/reversed.

4. Key Findings

#	Finding	Evidence	Defensive implication	Confidence
1	The exact hash is a Windows executable, not a document or script.	Hybrid Analysis metadata: `PE32 executable for MS Windows 4.00 (GUI), Intel i386, 6 sections`, MIME `application/vnd.microsoft.portable-executable`, size 107 KiB.	Prioritize endpoint, email attachment, web-download, and application-control telemetry.	High
2	Public reputation is malicious.	Hybrid Analysis verdict `malicious`; CrowdStrike Falcon static/ML verdict `malicious` with `100%`; MetaDefender 4/26 detections.	Block hash and investigate any endpoint execution.	Medium-high
3	The sample is likely packed or protected.	ESET label `Win32/Packed.PrivateEXEProtector.B suspicious application`; file size and PE characteristics are compatible with a small packed Windows loader.	Static string/import detection may be weak; require memory/process telemetry and unpacking if sample is acquired.	Medium
4	Midgate/Bifrose-style behavior is a plausible analytic hypothesis.	Lionic label `Trojan.Win32.Midgate.ljsQ`; Microsoft describes Midgate.A as carrying encrypted Bifrose and injecting	Hunt for suspicious `explorer.exe` injection, remote-access/backdoor behavior, credential-access indicators, and egress from shell	Low-to-medium for exact sample; medium for family context

		into `explorer.exe`; Kaspersky classifies Midgare as a trojan/spyware family.	processes.	
5	No public exact-sample behavioral report or relationships were available in accessible sources.	Hybrid Analysis `reports.count: 0`, `relations.displaySection : false`, no dropped/parent/bundled relations; Triage public page returned 404; Manalyzer reported no exact report.	Do not rely on network IOC absence; treat as a blind spot and run local sandbox/reversing if the binary is obtained.	Medium
6	Other matching samples should be handled as family-level, not sibling-level, evidence.	Kaspersky page lists example MD5s for Trojan.Win32.Midgare, but no exact relationship to the SHA256 was established.	Avoid merging IOC sets across unrelated packed samples without hash/import/config overlap.	Medium

Indicators of Compromise and observables:

Type	Value	Source	Handling
SHA256	`54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d`	Task input; Hybrid Analysis exact page	Block and hunt
File type	PE32 Windows GUI executable, Intel i386, six sections	Hybrid Analysis	Static triage pivot
Size	107 KiB	Hybrid Analysis	File inventory pivot
AV label	`Win/malicious_confidence_100%`	CrowdStrike Falcon via Hybrid Analysis	Reputation signal
AV label	`a variant of Win32/Packed.PrivateEXEProtector.B suspicious application`	ESET via MetaDefender/Hybrid Analysis	Packer/protector signal
AV label	`Trojan.Win32.Midgare.ljsQ`	Lionic via MetaDefender/Hybrid Analysis	Family hypothesis
AV label	`Trojan.Heur!.032100A1`	Gridinsoft via MetaDefender/Hybrid Analysis	Heuristic malware signal
Family examples, not confirmed siblings	`BA2A5CE93E9C6DB004E2385A21023906`, `599D526F7D92EFB90784C13415E6C39C`, `569457F4F1F31C3C76B2E51555052F9E`, `2F174BBFEE876790C821D6F1D33E7DAF`, `49C8320CA20DA845ECD6B`	Kaspersky Midgare family page	Research pivots only; do not block as related without local validation

	1D4BE679458`		
--	--------------	--	--

5. Threat Context

Midgare is an older Windows trojan naming family that appears in multiple vendor taxonomies. Kaspersky describes Trojan.Win32.Midgare as a malicious program designed to electronically spy on user activity, including keyboard input, screenshots, active-application lists, and exfiltration by channels such as email, FTP, or HTTP. Microsoft describes Trojan:Win32/Midgare.A as containing an encrypted copy of Backdoor:Win32/Bifrose and injecting the backdoor into the Windows shell process `explorer.exe` so the backdoor can run without being written separately to disk. These family descriptions are consistent with a malware class focused on stealthy execution, process injection, and remote-control or espionage outcomes.

The ESET PrivateEXEProtector label is also important. Private EXE Protector is a commercial executable-protection/packing technology, and packer labels are frequently not malware-family names by themselves. Packers can be used legitimately, but in malware triage they raise risk because they hide strings, imports, payloads, and control flow from simple static analysis. Dr.Web's PrivateExeProtector alias table shows broad association with backdoor, downloader, banker, autorun, and Bifrost/Bifrose-like detections across vendor labels, which supports the judgment that a PrivateEXEProtector-packed unknown PE should not be treated as benign based solely on missing behavioral telemetry.

For this exact SHA256, the absence of public sandbox telemetry is a blind spot rather than exculpatory evidence. A packed loader may fail to execute in unsupported sandbox environments, delay execution, require user interaction, or unpack only under host conditions. It may also be an uploaded file that has not been detonated publicly. Defender action should therefore center on containment and telemetry review rather than waiting for network IOCs.

MITRE ATT&CK mapping, scoped to observed and hypothesized behavior:

Tactic	Technique	Applicability	Evidence / rationale
Execution	T1204.002 User Execution: Malicious File	Likely	PE GUI executable typically requires user execution or delivery through another loader.
Execution	T1203 Exploitation for Client Execution	Family-level possible	Kaspersky Midgare page maps exploitation for client execution; no exact-sample exploit delivery was observed.
Defense Evasion	T1027 Obfuscated Files or Information	Likely	ESET packed/protector label and small PE loader profile.
Defense Evasion	T1055 Process Injection	Family-level possible	Microsoft Midgare.A description says Bifrose payload is injected into `explorer.exe`; not proven for this SHA256.
Defense Evasion	T1497.001 Virtualization/Sandbox Evasion: System Checks	Family-level possible	Kaspersky Midgare ATT&CK context includes virtualization/sandbox-evasion system checks.
Command and Control	T1071 Web Protocols	Family-level possible	Kaspersky describes exfiltration by HTTP among other channels for Midgare-like spyware behavior.
Collection	T1056.001 Keylogging / T1113	Family-level possible	Kaspersky describes keyboard

	Screen Capture		interception and screenshot capture for the family.
Exfiltration	T1041 Exfiltration Over C2 Channel	Family-level possible	Kaspersky family description includes sending collected information to the operator.

6. Detection and Hunting

Detection strategy should combine exact-hash blocking, reputation controls, packer/protector heuristics, and behavior hunts. The strongest exact observable is the SHA256; however, mature coverage should not stop there because packed backdoor loaders often mutate hashes while preserving execution traits such as shell-process injection, anomalous network traffic, and suspicious process access patterns.

Detection coverage matrix:

Control layer	Detection objective	Suggested logic	Priority
EDR / AV	Block exact SHA256 and alert on execution attempts.	Hash allow/block policy and custom IOC.	High
Email / web proxy	Prevent delivery or download of the hash.	Attachment/download hash matching; quarantine packed PE attachments from untrusted sources.	High
Endpoint behavior	Detect process injection into Windows shell processes.	Alert on unsigned/suspicious process opening `explorer.exe` with VM_WRITE/CREATE_THREADED style access and on abnormal child/remote thread behavior.	High
Endpoint inventory	Find small packed PE files in user-writable paths.	Hunt for recently created PE32 files around 107 KiB in Downloads, Temp, AppData, browser cache, and mail attachment directories.	Medium
Network	Detect unusual egress from `explorer.exe`.	Hunt for `explorer.exe` initiating external connections, especially to rare destinations or non-standard ports.	High
Identity / credential monitoring	Watch for follow-on credential theft or spyware behavior.	Correlate suspicious PE execution with credential-store access, browser credential reads, screenshots, or keyboard-hook API use.	Medium

Sigma rule for exact hash and packed/Midgate labeling in endpoint telemetry:

```

title: Malicious Packed PE Hash 54cf42a4 Midgate Hypothesis
id: 5ec1dbe6-5f04-4a25-86c7-54cf42a4ddf0
status: experimental
description: Detects execution or file creation for SHA256
54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d and adjacent packed/Midgate labels
when EDR telemetry preserves threat names.

```



```
author: Lost Boys Cyber
date: 2026-06-25
references:
-
https://www.hybrid-analysis.com/sample/54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d
logsource:
  product: windows
  category: process_creation
detection:
  selection_hash:
    Hashes|contains: 'SHA256=54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d'
  selection_threat_name:
    - ThreatName|contains: 'Midgare'
    - ThreatName|contains: 'PrivateEXEProtector'
    - ThreatName|contains: 'malicious_confidence_100'
  condition: selection_hash or selection_threat_name
fields:
- UtcTime
- Image
- CommandLine
- ParentImage
- User
- Hashes
- ThreatName
falsepositives:
- PrivateEXEProtector labels may occur on protected benign software; require reputation, source path,
signer, and user context review.
level: high
```

Microsoft Defender XDR hunt for exact hash execution or file presence:

```
let target_sha256 = "54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d";
DeviceFileEvents
| where SHA256 =~ target_sha256
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256, InitiatingProcessFileName,
InitiatingProcessCommandLine, InitiatingProcessAccountName
| union (
  DeviceProcessEvents
  | where SHA256 =~ target_sha256 or ProcessCommandLine has target_sha256
  | project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256,
InitiatingProcessFileName, ProcessCommandLine, AccountName
)
| order by Timestamp desc
```

Microsoft Defender XDR hunt for suspicious `explorer.exe` network egress after packed PE execution:

```
let lookback = 14d;
let suspicious_packed = DeviceProcessEvents
| where Timestamp > ago(lookback)
| where FileName endswith ".exe"
| where FolderPath has_any ("\\Downloads\\", "\\AppData\\", "\\Temp\\", "\\Users\\Public\\")
| where (SHA256 =~ "54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d" or
ProcessVersionInfoOriginalFileName == "" or ProcessVersionInfoCompanyName == "")
| project DeviceId, DeviceName, PackedTime=Timestamp, PackedImage=FileName, PackedPath=FolderPath,
PackedSHA256=SHA256, AccountName;
DeviceNetworkEvents
| where Timestamp > ago(lookback)
| where InitiatingProcessFileName =~ "explorer.exe"
| where RemoteIPType == "Public" or RemoteUrl != ""
| join kind=innerunique suspicious_packed on DeviceId
| where Timestamp between (PackedTime .. PackedTime + 6h)
| project Timestamp, DeviceName, AccountName, PackedTime, PackedImage, PackedPath, PackedSHA256,
InitiatingProcessFileName, RemoteIP, RemoteUrl, RemotePort, Protocol
| order by Timestamp desc
```

Splunk hunt for the exact hash and possible shell-process injection/network behavior:

```
index=edr sourcetype IN (sysmon:process, sysmon:file, sysmon:network, edr:telemetry)
(sha256="54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d" OR
```

```
hashes="*54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d*" OR threat_name="*Midgare*"
OR threat_name="*PrivateEXEProtector*")
| table _time host user parent_process process process_path command_line sha256 hashes threat_name
action
| sort - _time
```

YARA rule for triage repositories and quarantine stores:

```
rule LBC_Hash_54cf42a4_Packed_PE_Triage
{
  meta:
    author = "Lost Boys Cyber"
    date = "2026-06-25"
    description = "Triage rule for exact hash and suspicious small PE32 profile associated with
packed Midgare hypothesis"
    reference =
"https://www.hybrid-analysis.com/sample/54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97
d"
    sha256 = "54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d"
  condition:
    hash.sha256(0, filesize) == "54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d"
    or (
      uint16(0) == 0x5A4D and
      filesize < 200KB and
      pe.machine == pe.MACHINE_I386 and
      pe.number_of_sections == 6
    )
}
```

Triage notes for SOC analysts:

- Treat exact-hash hits as high severity until proven otherwise.
- For family-level heuristic hits, validate signer, prevalence, parent process, download source, path, command line, and outbound network behavior before containment decisions.
- If a host executed the file, collect memory or EDR process-injection telemetry before reboot when feasible.
- Look for `explorer.exe` egress and suspicious handles/remote-thread activity because the Midgare/Bifrose hypothesis specifically implies shell-process injection.
- Do not assume lack of public C2 IOCs means no network behavior; no public sandbox report was available for the exact sample.

7. Recommendations

Priority	Recommendation	Owner	Rationale
P0	Add SHA256 `54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d` to EDR, email, web proxy, and SIEM watchlists.	SOC / endpoint engineering	Exact public malicious verdict; low operational downside for blocking a known suspicious PE.
P0	Hunt last 30 days for file creation, process execution, and download events matching the hash.	SOC	Establish whether the observable has appeared in the environment.
P1	Hunt for `explorer.exe` outbound network connections and injection-adjacent events shortly after unknown packed PE execution.	Detection engineering	Covers Midgare/Bifrose-style behavior without relying on exact hash.
P1	If a copy is obtained, perform	Malware analysis	Needed to prove or reject the

	sample-backed analysis: static triage, packer identification/unpacking, TRASHcan detonation, Ghidra function review, strings/import recovery, and IOC extraction.		Midgare/Bifrose hypothesis for the exact binary.
P2	Add a temporary detection watch for PrivateEXEProtector-packed unsigned PE files launched from user-writable paths.	Detection engineering	Useful broad control, but higher false-positive risk from protected benignware.
P2	Preserve any related samples or detections as a small cluster only after validating hash, imphash, import table, section profile, config, or C2 overlap.	Threat intelligence	Prevents over-clustering based on generic AV labels.

Decision guidance:

- If this hash is found on a client endpoint: isolate the host if execution is confirmed; if only a blocked download exists, preserve the artifact/source URL and inspect user/browser/email telemetry.
- If only family-level Midgare detections are found: do not assume this exact sample is present; run the behavioral hunts and compare sample metadata before merging incidents.
- If the binary is recovered: prioritize unpacking and memory-backed analysis over superficial string review because PrivateEXEProtector labels indicate obfuscation/protection.

8. References

- [1] Hybrid Analysis / CrowdStrike Falcon Sandbox, exact sample page: [https://www.hybrid-analysis.com/sample/54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d`](https://www.hybrid-analysis.com/sample/54cf42a4ddf0e09c63643b95d0376a78680794649d9eaba5ab82077dbce7d97d)
- [2] Kaspersky Threats, Trojan.Win32.Midgare.ajzz: [https://threats.kaspersky.com/en/threat/Trojan.Win32.Midgare.ajzz/`](https://threats.kaspersky.com/en/threat/Trojan.Win32.Midgare.ajzz/)
- [3] Microsoft Security Intelligence search result, Trojan:Win32/Midgare.A: [https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Trojan:Win32/Midgare.A&threatId=-2147358090`](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Trojan:Win32/Midgare.A&threatId=-2147358090)
- [4] Dr.Web VMS search, Packer.PrivateExeProtector.A aliases: [https://vms.drweb.com/search/?q=Packer.PrivateExeProtector.A`](https://vms.drweb.com/search/?q=Packer.PrivateExeProtector.A)
- [5] MITRE ATT&CK, Process Injection T1055: [https://attack.mitre.org/techniques/T1055/`](https://attack.mitre.org/techniques/T1055/)
- [6] MITRE ATT&CK, Obfuscated Files or Information T1027: [https://attack.mitre.org/techniques/T1027/`](https://attack.mitre.org/techniques/T1027/)
- [7] MITRE ATT&CK, Virtualization/Sandbox Evasion T1497.001: [https://attack.mitre.org/techniques/T1497/001/`](https://attack.mitre.org/techniques/T1497/001/)

Source limitations and refresh note: this report should be refreshed if the sample is acquired, if a full VirusTotal/Triage/Hybrid sandbox report becomes available, or if any client telemetry produces concrete filenames, signer data, command lines, child processes, network destinations, registry keys, or dropped files.